OpenClaw 64GB VRAM tier: the multi-GPU sweet spot for 70B-class models

The 64GB VRAM tier — achieved through either 2× RTX 5090 32GB or 4× RTX 5060 Ti 16GB — unlocks a decisive upgrade over 48GB: 70B models at Q5_K_M quality with generous KV cache headroom, Nemotron-Super-49B at near-lossless Q8_0, and several new MoE models from late 2025 and early 2026. The upgrade from Q4_K_M to Q5_K_M on 70B-class models yields a measurable **~34% reduction in perplexity degradation** versus FP16, making this tier the practical ceiling for quality-conscious local 70B inference. Between the two GPU configurations, **2× RTX 5090 is roughly 1.5–2× faster** than 4× RTX 5060 Ti for the same model, thanks to 2× the aggregate memory bandwidth (3,584 vs. 1,792 GB/s). However, the 4-GPU configuration offers a compelling cost advantage at approximately half the price and enables simultaneous multi-model serving — a significant benefit for agentic workloads.

No genuinely new dense models in the 40B–120B total parameter range were released as open weights between November 2025 and February 14, 2026. The industry has shifted almost entirely to Mixture-of-Experts architectures, Runpod with new releases from major labs all exceeding 196B total parameters. The handful of new MoE models whose total parameter counts fall in the 40B–120B range — particularly Qwen3-Coder-Next (80B/3B active) MarkTechPost and Solar-Open-100B (102.6B/12B active) AI Models — fit comfortably in 64GB and bring genuinely strong capabilities, especially for coding and agentic tasks.

The new model landscape has gone full MoE

Between November 2025 and February 14, 2026, every major open-weight release from a leading lab used a Mixture-of-Experts architecture. The standout releases include **DeepSeek V3.2** (685B total/37B active, December 1, 2025), Galaxy **Mistral Large 3** (675B/41B active, December 2, 2025), Hugging Face IntuitionLabs **GLM-5** from Zhipu AI Bloomberg (744B/40B active, February 11, 2026), Medium +2 **Kimi K2.5** from Moonshot AI (1T/32B active, January 27, 2026), TechCrunch BentoML and **MiniMax M2.5** (230B/10B active, February 12, 2026). Hugging Face WinBuzzer None of these models fit in 64GB VRAM at any useful quantization level — their Q4_K_M sizes range from **~130 GB to over 600 GB**.

The models most relevant to the 64GB tier are the smaller MoE releases and coding specialists:

Qwen3-Coder-Next (80B total, 3B active, February 3, 2026) is the most exciting new arrival for this tier. This MoE model with 512 experts Clarifai MarkTechPost achieves **70.6% on SWE-bench Verified** using the SWE-Agent scaffold — competitive with models activating 10–20× more parameters. MarkTechPost At ~46 GB for Q4_K_M, it fits in 64GB with ample KV cache room and runs extremely fast due to its tiny 3B active parameter count. It ships with native tool calling support and an Apache 2.0 license. VentureBeat

Qwen3-Next-80B-A3B-Thinking Clarifai shares the same architecture Alibaba Cloud BentoML and fits identically in VRAM, but focuses on reasoning tasks, hitting **87.8% on AIME 2025** — outperforming the older Qwen3-32B thinking model. Clarifai

Devstral 2 (123B dense, December 2025) is the period's strongest coding model at **72.2% SWE-bench Verified**, Mistral AI Medium but its Q4_K_M weighs **74.9 GB** Hugging Face — exceeding 64GB. At Q3_K_M (60.6 GB), Hugging Face it barely fits with ~3 GB for KV cache, enough for only short contexts. Users running this model on 2× 5090 typically use ik_llama.cpp's graph split with custom IQ4 quantizations Hugging Face for a better quality-to-size tradeoff.

Solar-Open-100B (102.6B total, 12B active) from Upstage is a MoE model trained on 19.7 trillion tokens, AI Models with INT4 quantized versions around ~58 GB that likely fit at 64GB. It supports tool calling via a custom parser and targets reasoning workloads, AI Models though public benchmark data remains limited.

Several other new releases were confirmed to **not** be released or to fall outside the target range: **DeepSeek V4** is expected around February 17, 2026 but is not yet available. WaveSpeedAI Introl **Llama 4 Behemoth** (~2T/288B active) remains in training. DataCamp Wikipedia No new Llama 4 variants were released beyond Scout and Maverick Hugging Face from April 2025. Wikipedia Contabo **Nemotron 3 Super** (~100B/10B active) was announced but won't ship until H1 2026. NVIDIA Newsroom The New Stack No Qwen3 dense model larger than 32B exists. No Phi-4 model exceeds 14B. Microsoft No Falcon 3 model exceeds 10B. AWS No Gemma 3 model exceeds 27B. Contabo Google

Complete model roster for 64GB VRAM

The table below shows every model that fits fully in 64GB VRAM at its optimal quantization level, ranked by the recommended quantization quality. The "KV headroom" column estimates remaining VRAM after weights are loaded, available for KV cache and CUDA overhead.

Model	Params	Architecture	Quant	GGUF size	KV headroom	Context capacity	Tool calling
Qwen3-Coder-Next 80B	80B (3B active)	MoE	Q4_K_M	~46 GB	~18 GB	128K+	✅ Native
Qwen3-Next-80B-A3B Thinking	80B (3B active)	MoE	Q4_K_M	~46 GB	~18 GB	128K+	✅
Llama 3.3 70B Instruct	70B (dense)	Dense	Q5_K_M	49.9 GB	~10–14 GB	16–32K	✅
DeepSeek-R1-Distill-Llama-70B	70B (dense)	Dense	Q5_K_M	49.9 GB	~10–14 GB	16–32K	Limited
Nemotron-Super-49B v1.5	49B (dense)	Dense (NAS)	**Q8_0**	53 GB	~8–11 GB	8–16K	✅
Qwen2.5-72B Instruct	72B (dense)	Dense	Q5_K_M	54.5 GB	~5.5–9.5 GB	8–16K	✅
Solar-Open-100B	102.6B (12B active)	MoE	Q4/INT4	~58 GB	~6 GB	8K	✅ Custom
Llama 4 Scout	109B (17B active)	MoE	Q3_K_M	51.8–54.3 GB	~10–12 GB	16–32K	✅
Devstral 2 123B	123B (dense)	Dense	Q3_K_M	60.6 GB	~3 GB	2–4K only	✅ Native

Nemotron-Super-49B at Q8_0 deserves special attention. At 53 GB, this model runs at near-lossless quality (Hugging Face) — Q8_0 introduces less than 2% perplexity increase versus FP16, making it effectively indistinguishable from the original weights for most tasks. (Hugging Face) No other model at 64GB can run at such high quantization fidelity. For general-purpose agentic work where quality-per-bit matters most, **Nemotron-Super-49B at Q8_0 is arguably the single best model at this tier**.

For coding-focused agentic workflows, the picture is different. **Qwen3-Coder-Next at Q4_K_M** offers the best combination of SWE-bench performance (70.6%), (Qwen) VRAM efficiency (46 GB), inference speed (only 3B active params), and native tool calling — making it the clear pick for OpenClaw's coding agent use case. Its MoE architecture means it generates tokens far faster than dense 70B models (NVIDIA) while fitting comfortably in 64GB. (Hugging Face)

The models that **do not** fit at 64GB at Q4_K_M, confirmed with exact GGUF file sizes: GPT-oss-120B (**87.9 GB**, (Hugging Face) with MoE weights stuck at MXFP4 preventing meaningful compression), (Hugging Face +4)

Command R+ 104B (Hugging Face) (**62.8 GB** (Hugging Face) — tantalizingly close but leaves zero KV cache room), (Hugging Face) Mixtral 8x22B (**85.6 GB**), (Hugging Face) and DBRX (**~75 GB**). (huggingface)

Why 64GB matters: the Q5_K_M quality leap over 48GB

The primary value proposition of 64GB over 48GB is not access to entirely new model classes — it is **higher quantization quality on 70B models and larger context windows**. At 48GB, a Llama 3.3 70B at Q4_K_M (42.5 GB) fits with modest KV cache room. At 64GB, the same model at Q5_K_M (49.9 GB) fits (Hugging Face) with 10–14 GB of headroom — enough for **16K–32K context** versus the 8K practical limit at 48GB.

The quality difference between Q4_K_M and Q5_K_M is measurable. Based on llama.cpp perplexity benchmarks, **Q4_K_M adds +0.0532 perplexity** versus FP16, while **Q5_K_M adds only +0.0353** — a 34% reduction in quality degradation. Community blind testing suggests this gap is noticeable primarily in precision-heavy tasks: mathematical reasoning, structured output generation, and multi-step code refactoring. For conversational tasks, the difference is subtle. For agentic coding workflows where tool calling precision matters, Q5_K_M provides a meaningful edge.

The 64GB tier also uniquely enables:

- **Nemotron-Super-49B at Q8_0** (53 GB) — impossible at 48GB where only Q6_K (40.9 GB) fits (Hugging Face)
- **Qwen2.5-72B at Q5_K_M** (54.5 GB) — at 48GB, this model is limited to Q4_K_M (47.4 GB) (Hugging Face)
- **Llama 4 Scout at Q3_K_M** (~52–54 GB) (Hugging Face) — a 109B MoE model (Maximilian-schwarzmueller) with 10M context capability (Hugging Face +3)
- **Future-proofing**: model sizes are trending upward, and the 64GB tier provides approximately 2 years of headroom before the next VRAM threshold becomes necessary

The **diminishing returns argument** has merit. The jump from 24GB → 48GB is transformative (it enables 70B models entirely). (Arsturn) The jump from 48GB → 64GB is incremental but valuable — better quant quality, larger context windows, and access to ~100B MoE models. Community testing of Qwen 110B on 2× RTX 5090 showed only **7.22 t/s** at 20% GPU utilization, (Database Mart) confirming that 64GB enables these models but doesn't make them fast. The strongest case for 64GB is KV cache headroom: long-context agentic workloads on 70B models can consume 10–15 GB of KV cache, (Localllm) and 64GB avoids the cliff where performance degrades catastrophically.

Two RTX 5090s versus four RTX 5060 Ti cards

Both configurations deliver 64GB total VRAM but with dramatically different performance profiles. The comparison hinges on memory bandwidth, which governs token generation speed for autoregressive LLM inference.

Metric	2× RTX 5090	4× RTX 5060 Ti 16GB
Total VRAM	64 GB	64 GB
Aggregate bandwidth	**3,584 GB/s**	**1,792 GB/s**
Total CUDA cores	43,520	18,432
PCIe interface	2× PCIe 5.0 x16	4× PCIe 5.0 x8
NVLink	Not available	Not available
Total TDP	1,150W	720W
Approximate cost	~$4,000	~$1,720–$2,600

Confirmed benchmarks: Llama 3.3 70B on 2× RTX 5090 achieves **~27 tokens/sec** with Ollama's default pipeline parallelism — matching H100 speeds at roughly 25% of the cost. (Database Mart) With ik_llama.cpp's graph-split tensor parallelism, this could reach an estimated **40–50 t/s** based on the 3–4× speedup that tool provides over pipeline parallelism. (Medium) Neither GPU supports NVLink (discontinued on consumer GeForce after RTX 3090), so all inter-GPU communication travels over PCIe 5.0. (Best GPUs for AI)

For 4× RTX 5060 Ti, no direct 70B benchmarks exist, but extrapolation from dual 5060 Ti data (Qwen3 32B Q4 at 18.2 t/s on 2× cards) (hardware-corner) and the bandwidth ratio suggests **~12–18 t/s with pipeline parallelism** and **~20–25 t/s with tensor parallelism** for a 70B Q4_K_M model. The 2× RTX 5090 configuration is approximately **1.5–2× faster** for single-model inference.

The 4× RTX 5060 Ti configuration has one decisive advantage: **multi-model serving**. With 4 GPUs, you can allocate 2 GPUs per model and run two separate ~30B models simultaneously — ideal for agentic architectures where a fast "router" model and a powerful "worker" model operate concurrently. The 2× RTX 5090 can also split models (one per 32GB card), but with fewer permutations. For OpenClaw's agent framework, the ability to run Qwen3-Coder-Next (46 GB across all 4 GPUs) alongside a smaller tool-calling router model could be architecturally significant.

For agentic workloads, community consensus suggests **25–30 t/s minimum** for comfortable code generation and **≥50 t/s** for fluid "vibe coding." (Simon Willison) (Shir-man) Time to First Token (TTFT) matters more than raw generation speed for agents that send long contexts but receive short outputs. (LanceDB) The 2× RTX 5090 configuration meets the 25–30 t/s threshold easily; the 4× RTX 5060 Ti configuration likely falls short for dense 70B models but excels with the new lightweight MoE models (Qwen3-Coder-Next at only 3B active params would generate tokens extremely fast).

The inference toolchain for consumer multi-GPU

The multi-GPU inference landscape has matured significantly, with **ik_llama.cpp** emerging as the breakthrough

tool for consumer setups. Here is how each major tool handles 2-GPU and 4-GPU configurations:

ik_llama.cpp (the recommended default) introduced **split mode graph** (`-sm graph`), implementing tensor parallelism at the GGML graph level (Medium) using NVIDIA NCCL for topology-aware communication. (Medium) This keeps all GPUs at 100% utilization (Medium) — a stark contrast to pipeline parallelism where GPUs idle 75% of the time in a 4-way split. (ROCm Blogs) Community reports show **3–4× performance gains** over standard llama.cpp's layer split. (Medium) Configuration is straightforward: `llama-cli -m model.gguf -sm graph -ngl 99`. (Medium) It supports 4+ GPUs and works with P2P GPU memory access over PCIe 5.0.

vLLM provides production-grade tensor parallelism and is the best option for serving workloads with concurrent requests. For consumer GPUs without NVLink, the vLLM documentation explicitly recommends **pipeline parallelism over tensor parallelism** (`--pipeline-parallel-size 4`), as the AllReduce overhead of TP across PCIe is substantial. (vLLM) (vLLM) vLLM's real strength is batched inference — multiple agent queries running simultaneously.

Standard llama.cpp and **Ollama** default to pipeline parallelism (layer split), which provides **no speedup for token generation** — only one GPU computes at a time. Their primary value is VRAM pooling to fit larger models. Ollama auto-detects multiple GPUs and distributes layers automatically, (Boredconsultant) making it the easiest option for casual use, but it is not suitable for performance-critical workloads.

ExLlamaV2/V3 supports tensor parallelism (Medium) via `--gpu_split auto` with reported 33%+ throughput increases. (GitHub) ExLlamaV3 adds expert parallelism for MoE models, distributing experts across GPUs. (DeepWiki) **TabbyAPI**, the official API server for both ExLlama versions, (GitHub) inherits full multi-GPU support via `gpu_split_auto: True` in its config. (GitHub)

SGLang implements the most complete parallelism stack: TP, PP, expert parallelism, and data parallelism. (DeepWiki) Its pipeline parallelism achieves over **80% strong scaling efficiency** at PP=4. (LMSYS Org) **KoboldCpp** supports 4-GPU via layer split only (no tensor parallelism). (GitHub) **TensorRT-LLM** supports consumer GPUs including RTX 50-series (Jan) with FP4/NVFP4 (GitHub) (NVIDIA Developer) and is **62.6% faster** than llama.cpp on single-GPU benchmarks, (Jan) but multi-GPU setup complexity is high.

For the OpenClaw framework, the recommended stack is **ik_llama.cpp with `-sm graph`** for GGUF models requiring maximum single-user speed, and **vLLM with PP=4** for serving multiple concurrent agent requests.

Recommended configurations for OpenClaw at 64GB

For 2× RTX 5090 (performance-optimized): The optimal single-model choice is **Llama 3.3 70B at Q5_K_M** (Elephas) (49.9 GB) using ik_llama.cpp's graph split, delivering an estimated 40–50 t/s with high quantization fidelity and 10–14 GB for KV cache. For coding-focused agents, **Qwen3-Coder-Next 80B** at Q4_K_M (46 GB) offers superior SWE-bench scores with even faster generation thanks to its 3B active parameter MoE architecture. For maximum per-token quality, **Nemotron-Super-49B at Q8_0** (53 GB) provides near-lossless inference. (Hugging Face) A dual-model architecture could pair Qwen3-Coder-Next (loaded into ~46 GB of shared VRAM) with a smaller tool-calling router.

For 4× RTX 5060 Ti (cost-optimized): The same models fit, but dense 70B models will run at roughly half the speed (~20–25 t/s with TP). The MoE models shine here: Qwen3-Coder-Next at 3B active params will generate tokens extremely fast even on slower individual cards. The unique advantage is **simultaneous multi-model serving**: allocate 2× 5060 Ti (32 GB) to a 70B model at Q4_K_M while the other 2× 5060 Ti run a 30B routing/planning model — enabling a true multi-agent architecture without model swapping overhead.

For coding and tool calling specifically, the strongest new options are Qwen3-Coder-Next (70.6% SWE-bench, native tool calling, Apache 2.0, Qwen 46 GB at Q4), followed by Devstral 2 at Q3_K_M (72.2% SWE-bench, Medium 60.6 GB — barely fits with short context). GPT-oss-120B remains the best tool-calling model in the 100B+ class Clarifai +2 (strong TauBench scores, native function calling) OpenAI +3 but does not fit at 64GB due to its unusual MXFP4 quantization that prevents further compression. Hugging Face

Conclusion

The 64GB tier represents a **meaningful but not transformative** upgrade over 48GB. Its core value lies in three areas: upgrading 70B models from Q4_K_M to Q5_K_M (a genuine quality improvement for precision-sensitive agentic tasks), enabling Nemotron-Super-49B at near-lossless Q8_0 (the highest-fidelity single model at this tier), and providing KV cache headroom for longer context windows on 70B models. The new MoE models from late 2025 and early 2026 — particularly **Qwen3-Coder-Next** — are the most significant additions, delivering frontier-class coding performance in a 46 GB footprint Marc0 with blazing-fast inference speeds.

The elephant in the room is that the massive new MoE models (DeepSeek V3.2, Mistral Large 3, GLM-5, Kimi K2.5, MiniMax M2.5) all require **400+ GB** and remain completely out of reach for 64GB consumer configurations. The 64GB tier is definitively the domain of 70B-class dense models and sub-120B MoE models — not the frontier MoE generation. For OpenClaw's architecture, the practical recommendation is **ik_llama.cpp with graph-split tensor parallelism** running either Qwen3-Coder-Next (for coding/tool-calling agents) or Llama 3.3 70B at Q5_K_M (for general-purpose agents), with DeepSeek V4's imminent release Introl potentially reshaping this landscape within days. WaveSpeedAI Ai-supremacy